

# SZL Holdings — Audit Delta (v2)

## What changed between the May 4 dossier (v1) and the May 5 refresh (v2)

This document records the concrete state changes made between 2026-05-04 (the v1 dossier publish date) and 2026-05-05 (the v2 dossier publish date), prepared for the May 6, 2026 Empire APEX session with counselor Mercy McInnis.

### 1. New peer-style papers published

**v9 — *The Lutar Invariant Family* v1 → v7 → Ω: From Three-Term Foundation to Bianchi-Closed Fiber Bundle**

| Field          | Value                                                             |
|----------------|-------------------------------------------------------------------|
| Repo           | github.com/szl-holdings/ouroboros-thesis                          |
| Tag            | paper-v9-1.0.0                                                    |
| Length         | 17 pages                                                          |
| File size      | 234 KB PDF                                                        |
| Author         | Stephen P. Lutar (SZL Holdings) — ORCID 0009-0001-0110-4173       |
| Date           | 2026-05-05                                                        |
| Build pipeline | markdown-it@14.1.1 → Chromium PDF (system Chromium at NixOS path) |
| Zenodo deposit | Re-fired by tag re-creation; release ID 317669393                 |

**v10 — The Audit Closure Operator  $\Lambda_{10}$ : Formalising the Implementation Contract of the Lutar Family**

| Field          | Value                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Repo           | same                                                                                                                                       |
| Tag            | paper-v10-1.0.0                                                                                                                            |
| Length         | 11 pages + Appendix A essay + Lutar-family one-pager                                                                                       |
| File size      | 211 KB PDF                                                                                                                                 |
| Date           | 2026-05-05                                                                                                                                 |
| Codex schema   | alloy.supreme_knowledge/v11-UNIFIED-OPERATIONAL (76 nodes, 95 edges; v10 adds the lutar_v10 node and a single derives edge from lutar_v7 ) |
| Live binding   | POST /api/ouroboros/lutar/v10 runs lutarV10Audit() against the live shipping repo on every test run                                        |
| Zenodo deposit | Re-fired by tag re-creation; release ID 317669402                                                                                          |

**What v10 does (worth landing in conversation)**

v10 introduces no new physical L-term. Its only purpose is to certify, layer by layer, that every formula in v9 actually executes against the live shipping repo. v10 is a meta-invariant on v9.  
**The platform audits its own thesis.**

**2. New Alloy operator surfaces**

The following pages were added to Alloy in the May 4–5 cycle. All are referenced in the v2 demo guide.

| Page                   | Path                       | Purpose                                                                    |
|------------------------|----------------------------|----------------------------------------------------------------------------|
| Trust Center           | /trust-center              | Constitutional surface — proof, covenants, attestation                     |
| Trust Exchange         | /trust-exchange            | Proof-distribution surface (regulators, auditors, partners)                |
| Public Trust Portal    | /public-trust-portal       | Externally-facing proof packets                                            |
| Agent Zero Trust       | /agent-zero-trust          | Runtime policy gate for every agent action                                 |
| Argo (decision engine) | /alloy/argo + /argo-bridge | Champion policies, mirror eval, counterfactuals, reward-hacking guardrails |
| ArgoForge              | /alloy/argo-forge          | Champion-policy authoring                                                  |
| MirrorEval             | /alloy/mirror-eval         | Off-policy evaluation                                                      |
| Counterfactuals        | /alloy/counterfactuals     | Counterfactual rollout analysis                                            |
| RewardHacking          | /alloy/reward-hacking      | Guardrail surface against reward hacking                                   |
| PSYCHE                 | (observatory)              | Emergent-sentience observatory                                             |

## Argo regressions resolved in the same cycle

- **SelfPlayArena replay playback** — `setInterval` driven, 900 ms per step, clears on pause/unmount, auto-stops at the last frame.
- **SelfPlayArena recency sort** — now timestamp-based via `lastMatchAt: string` ISO field; `Date.getTime()` comparison instead of id-string compare.
- **Argo event timestamps** — `Date.now()` removed from generated events; replaced with fixed anchor constant `ANCHOR_MS = 1746403200000` (2026-05-05T00:00:00Z) so all 90 generated event timestamps are fully deterministic.
- **HomePage Argo section** — collapsed to a single card with three KPIs (Champion Policies: 6, World-Model Acc: 89.1%, Throughput: 31.4 ev/s) + Bridge link.
- **Bridge cross-links** — Reward Hacking, Mirror Eval, and Counterfactuals added to the cross-link footer bar.

## 3. New customer-facing artifact: ROSIE — Unified Decision Fabric

A new artifact was scaffolded and registered:

- **Slug:** `/rosie/`
- **Display name:** ROSIE — Unified Decision Fabric

- **Pages:** Identity, Optimizer, Fabric, Research, Proof, Evidence Bench (six pages, hash-routed)
- **Role:** Operator surface for CPS payloads — execute, watch proof packets emit, approve or roll back
- **Workflow:** `artifacts/rosie: web` (registered with the artifact system)

This brings the customer-facing surface count from 7 → **8**. Verified-numbers tables across all v2 documents reflect this.

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## 4. Covenant Proof Standard (CPS) — first-class API shipped

| Endpoint                                         | Purpose                                               |
|--------------------------------------------------|-------------------------------------------------------|
| <code>GET /api/cps/payloads</code>               | List registered payloads                              |
| <code>GET /api/cps/payloads/:id</code>           | Resolve a payload definition                          |
| <code>POST /api/cps/runs</code>                  | Execute a payload run                                 |
| <code>GET /api/cps/runs/:id</code>               | Inspect a run, its proof receipts, and approval state |
| <code>POST /api/cps/runs/:id/approve</code>      | Approve a gated step at the caller's tier             |
| <code>POST /api/cps/runs/:id/rollback</code>     | Roll a completed run back to a prior verified state   |
| <code>POST /api/cps/payloads/:id/maturity</code> | Promote/demote a payload's maturity mode              |
| <code>GET /api/cps/runs</code>                   | List runs                                             |
| <code>GET /api/cps/approvals</code>              | List pending approvals                                |

Implementation lives at `artifacts/api-server/src/lib/domain-services/cps/`. Three flagship payloads ship pre-registered. Per-lane payloads are rolling out.

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## 5. Live agent gateway service

Workflow `artifacts/api-server: agent-gateway` is now registered alongside the main `api-server` workflow. The gateway sits in front of every agent action and enforces OPA bundle policy at the runtime boundary. Agent Zero Trust ( `/agent-zero-trust` ) is the operator-facing surface for that gate.

## 6. Investor zoom-out audit pass — Series A polish

A skeptical-investor audit was walked across all eight artifacts on 2026-05-05. Per-artifact reports landed at `docs/audits/<artifact>.md` and a canonical 5-minute investor demo path was committed to `docs/audits/INVESTOR_DEMO_PATH.md`. Targeted Tier-1 fixes applied this pass (highest-confidence, lowest-risk):

| Artifact   | Fix                                                                                                                                                                                                                                       |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conduit    | Throughput chart was on a hardcoded <code>Date.parse('2026-05-05T03:55:00Z')</code> “now” — replaced with <code>Date.now()</code> so the rolling 12-hour window never freezes.                                                            |
| Sentra     | Removed synthetic <code>proof_id: 'proof-s6-TBD'</code> placeholder on the pending step. Silenced <code>[sentra] fabric proof emission failed console.warn</code> with documented best-effort contract.                                   |
| Counsel    | Removed synthetic “SEC Filing Deadline — Global Operations” injection from the Risk Exposure Desk fallback. Empty-state now renders honestly.                                                                                             |
| Terra      | Removed <code>mockConfidence</code> / <code>mockEscalation</code> columns from the Property Detail tenant table. Banner stacking on narrow viewports fixed.                                                                               |
| Vessels    | Removed personal Medium link from marketing footer. Replaced “Updated 12s ago” with “Live AIS feed”. Stripped <code>[DEMO]</code> prefix from <code>AtelierSpaceEmbed</code> fallback transcript.                                         |
| Carlota Jo | Command palette rewired to <code>BASE_URL</code> -aware <code>navTo()</code> across all 23 commands. Removed invented “847 enterprise contracts” example exchange. Replaced intake <code>timeline: 'TBD'</code> default with honest copy. |
| Alloy      | Strategy → Governance / Team / Fabric: TBD placeholders purged. Doctrine fallback URL <code>sentinel-sr.example</code> replaced with the real <code>alloy.szlholdings.com/doctrines/sentinel-sr</code> .                                  |

The pass was explicit about what it did **not** do: full visual-token harmonization and a wholesale copy rewrite across all eight artifacts. Those are the scope of the queued downstream “GitHub org pristine pass — Fortune 500 / Series A polish” task.

## 7. Test coverage expansion

- New file: `artifacts/api-server/src/routes/__tests__/lyte-signals-incidents-playbooks.test.ts` — 19/19 passing. Mirrors the `lyte-action-queue` mock pattern; covers `GET/POST /lyte/signals` pagination + create + `broadcastWs`, `PATCH /lyte/incidents/:id`, `PATCH /lyte/signals/:id` 404.
- Live LaaS contract test suite *Lutar v10 — exhaustive-audit* is wired to `lutarV10Audit()` and runs against the shipping repo on every test run.

## 8. Files added or changed in this delta window

```
docs/thesis/v9-canonical.md          (new)
docs/thesis/v10-canonical.md         (new)
docs/thesis/v10-essay.md             (new – Appendix A)
docs/thesis/v10-onepager.md          (new – Lutar-family one-pager)
.local/deliverables/ouroboros-thesis-v9.pdf (built artifact)
.local/deliverables/ouroboros-thesis-v10.pdf (built artifact)
docs/audits/alloy.md                 (new)
docs/audits/conduit.md               (new)
docs/audits/sentra.md                (new)
docs/audits/counsel.md               (new)
docs/audits/terra.md                 (new)
docs/audits/carlota-jo.md             (new)
docs/audits/vessels.md               (new)
docs/audits/INVESTOR_DEMO_PATH.md    (new)
docs/audits/README.md                (new – tracking index)
artifacts/rosie/...                  (new artifact: 6 pages)
artifacts/api-server/src/routes/cps/index.ts (new)
artifacts/api-server/src/lib/domain-services/cps/... (new)
artifacts/api-server/src/lib/domain-services/cps/payloads.ts (3 flagship payloads)
artifacts/alloy/src/pages/argo/... (new)
artifacts/alloy/src/pages/{TrustCenter,TrustExchange,
  PublicTrustPortal,AgentZeroTrust,ArgoForge,
  MirrorEval,Counterfactuals,RewardHacking}.tsx (new)
artifacts/conduit/src/.../dashboard.tsx (Date.now() fix)
artifacts/sentra/src/.../governed-adversary-loop.tsx (proof-id placeholder removed)
artifacts/vessels/src/.../MarketingFooter.tsx (Medium link removed)
artifacts/vessels/src/.../marketing-home.tsx (Live AIS feed)
artifacts/terra/src/.../property-detail tenant table (mock columns removed)
artifacts/terra/src/App.tsx (banner stacking fix)
artifacts/counsel/src/.../risk-exposure-desk.tsx (synthetic SEC injection removed)
artifacts/carlota-jo/src/...command-palette + 23 commands (BASE_URL nav)
artifacts/api-server/start.sh (agent-gateway sidecar wiring)
```

Remote pushes:

- `szl-holdings/ouroboros-thesis@papers/v9/` — v9 + v10 PDFs deposited (commit `dd6c01d6`).
- Tags `paper-v9-1.0.0` and `paper-v10-1.0.0` recreated to re-fire Zenodo webhook with PDFs in source archive.
- Releases `317669393` (v9) and `317669402` (v10) recreated.

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## 9. What is still outstanding

The downstream “GitHub org pristine pass — Fortune 500 / Series A polish” task ( `#4753` ) covers the Tier-2/Tier-3 work surfaced by the per-artifact audit reports — visual-token harmonization across artifacts and the cross-artifact copy rewrite. Two specific remaining Tier-1-ish items: `SZL Holdings — Audit Delta May 5 · SZL Holdings · Confidential — Empire APEX`

1. **Counsel** — wire the missing tooltip provider for the `content` prop usage in `risk-exposure-desk.tsx:236`.
2. **Carlota Jo** — resolve “Carlota Jo” vs “Rosa” persona/identity drift across pages (deferred to the copy-rewrite pass to avoid creating new inconsistency in isolation).

Forged-author commits in `ouroboros-thesis` git history (four commits listing the operator’s email as author — `29de9e27b`, `2159e47aa`, `2fa3a6e50`, `84fbd4eac` ) still require a local `git filter-repo` followed by force-push from the operator’s local clone. The platform sandbox blocks destructive git operations.

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End of delta.